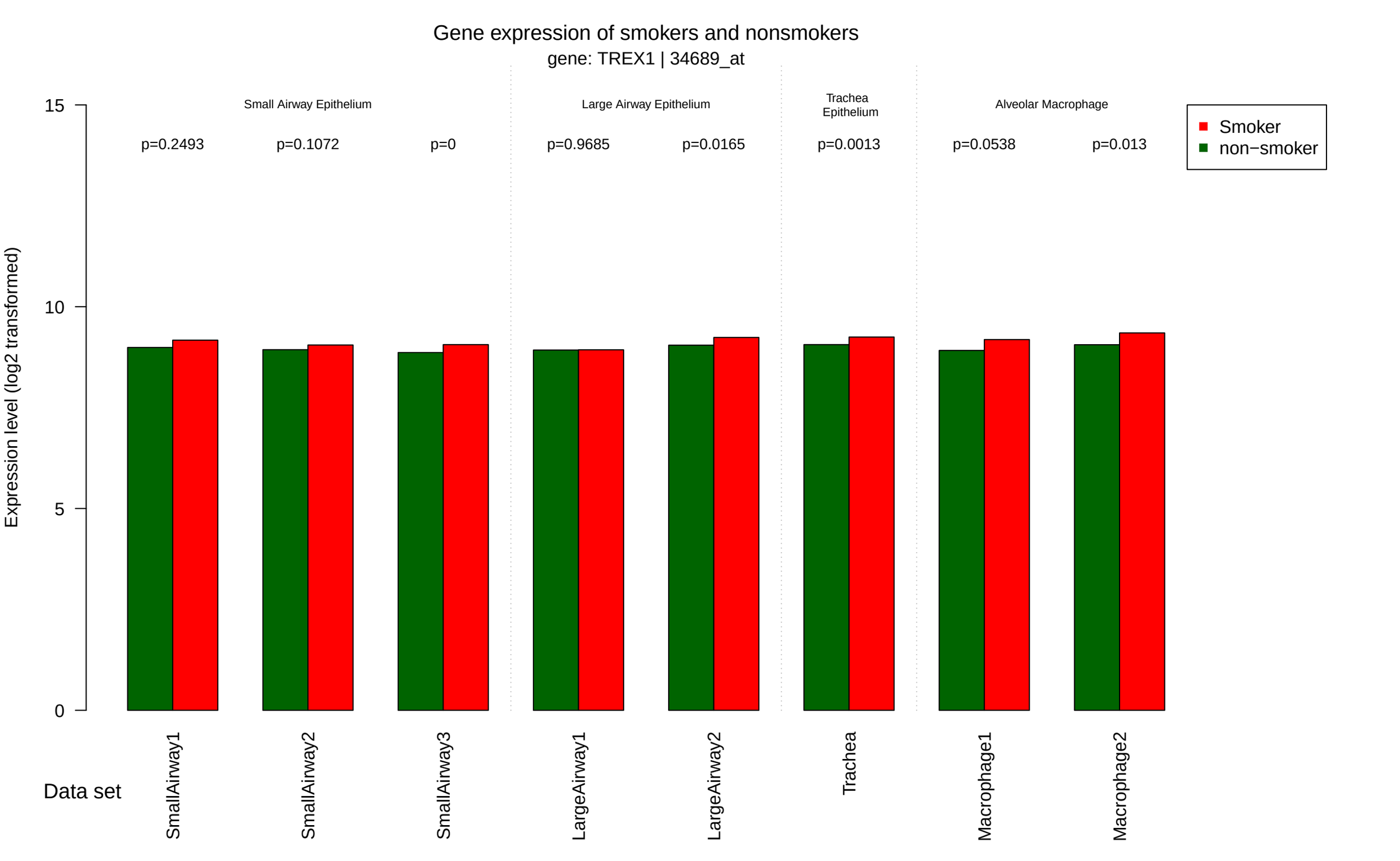
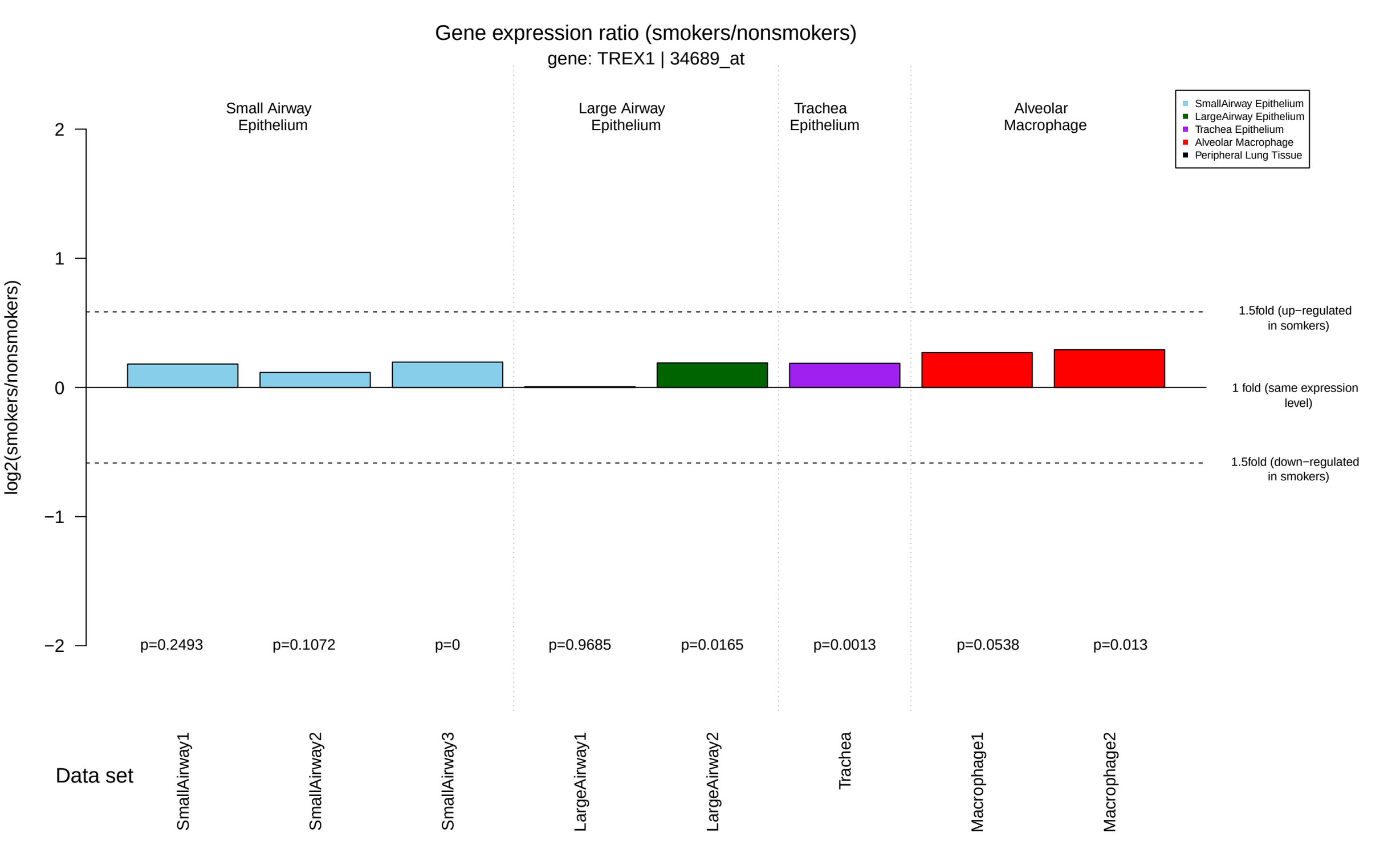
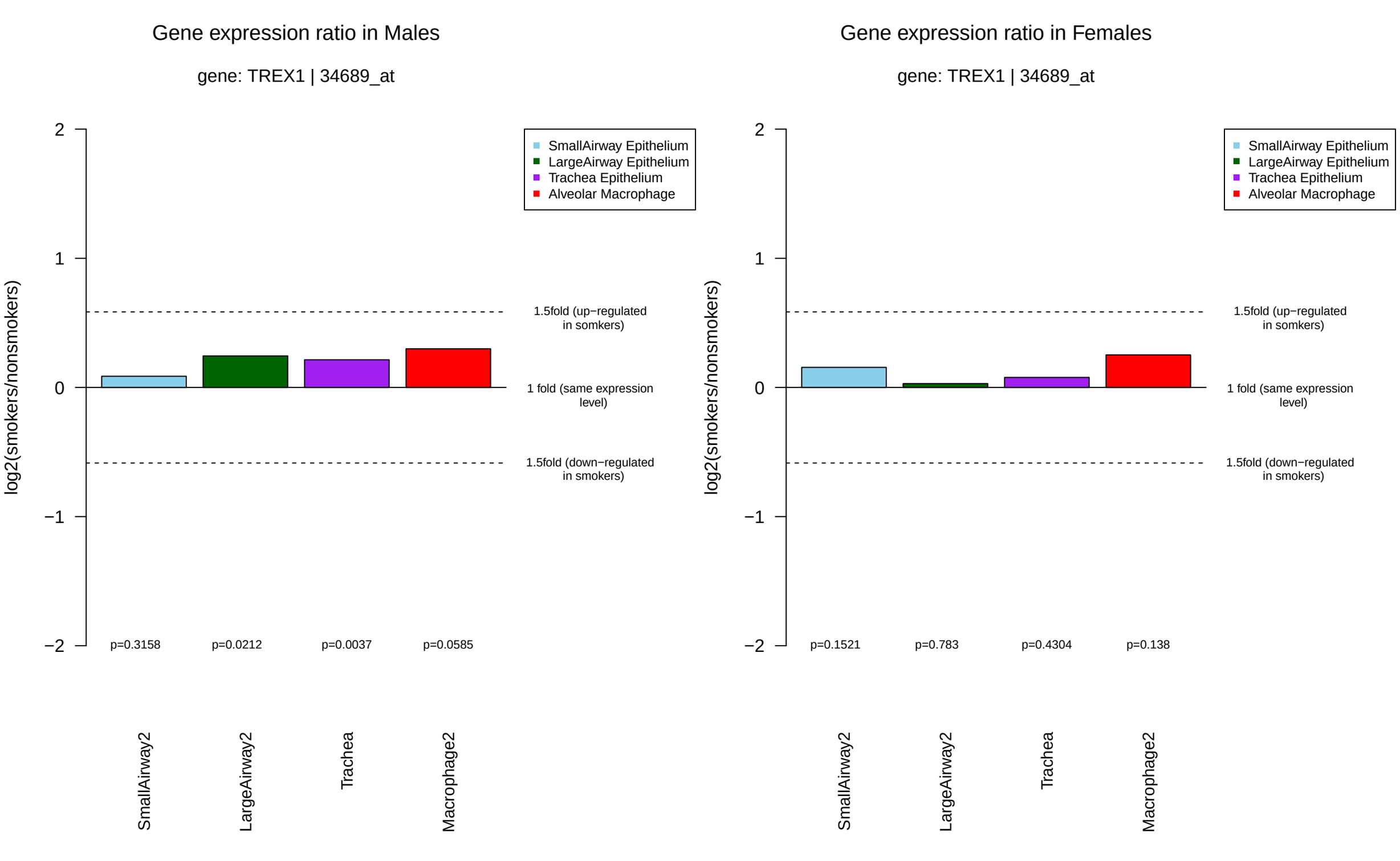
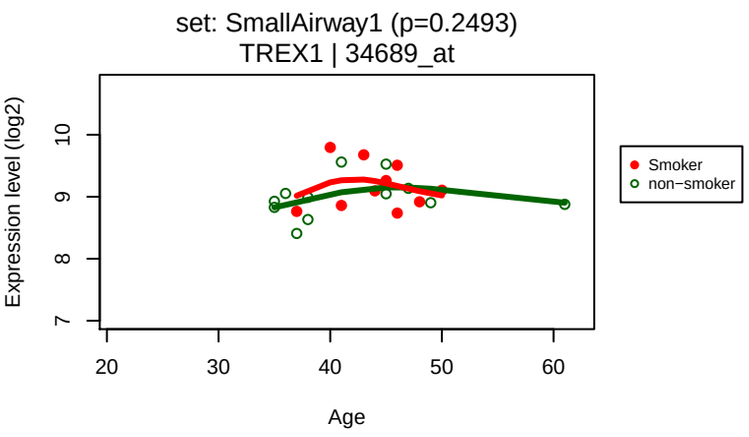
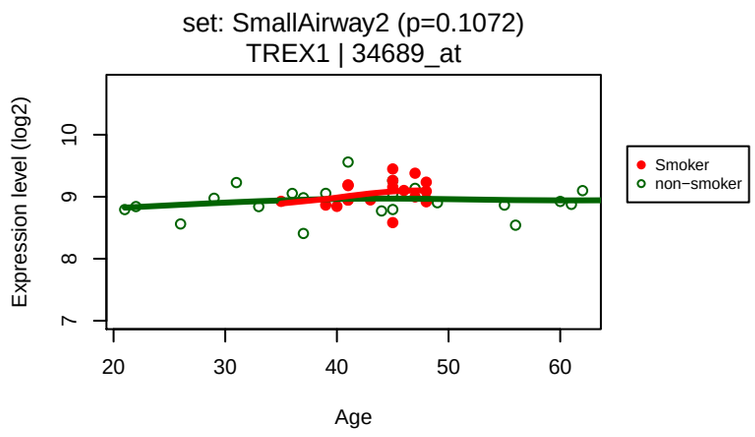
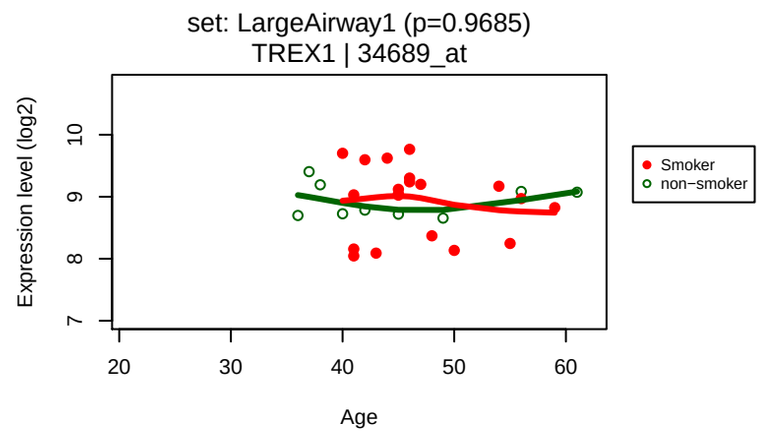

Gene expression of smokers and nonsmokers
gene: TREX1 | 34689_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2493
p=0.1072
p=0
p=0.9685
p=0.0165
p=0.0013
p=0.0538
p=0.013
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

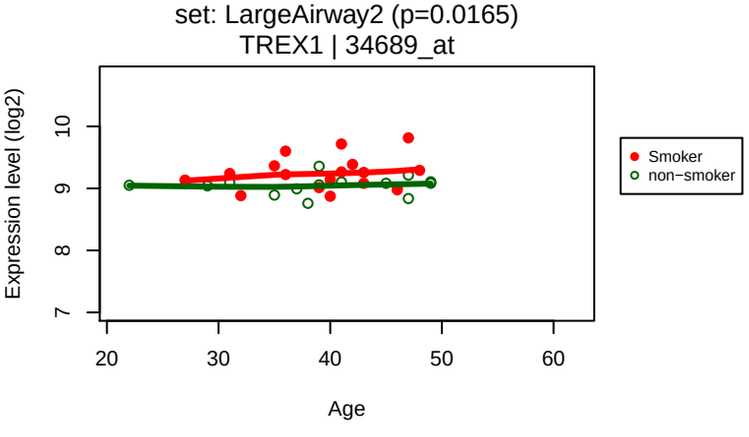

Gene expression ratio (smokers/nonsmokers)
gene: TREX1 | 34689_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2493
p=0.1072
p=0
p=0.9685
p=0.0165
p=0.0013
p=0.0538
p=0.013
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

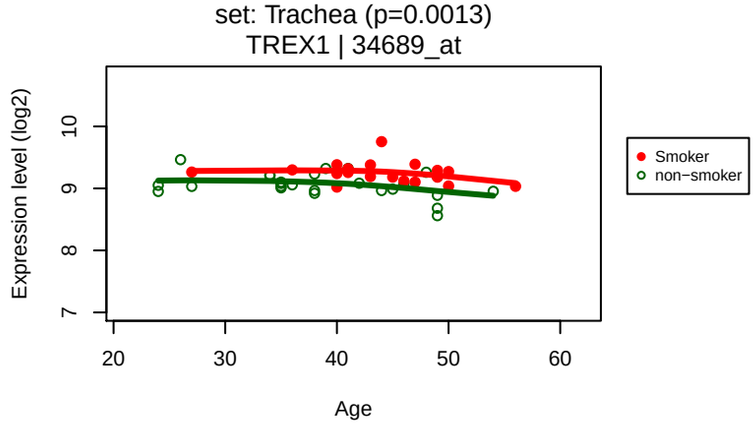

Gene expression ratio in Males
gene: TREX1 | 34689_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3158
p=0.0212
p=0.0037
p=0.0585
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TREX1 | 34689_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1521
p=0.783
p=0.4304
p=0.138
SmallAirway2
LargeAirway2
Trachea
Macrophage2

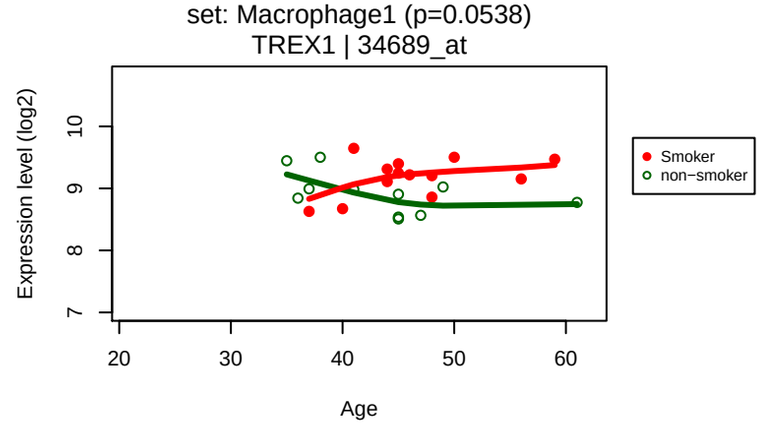

set: SmallAirway1 (p=0.2493)
TREX1 | 34689_at
Expression level (log2)
Age
Smoker
non−smoker

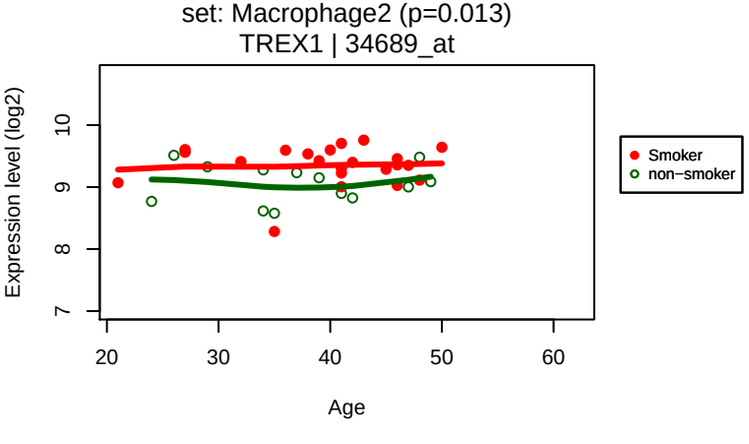

set: SmallAirway2 (p=0.1072)
TREX1 | 34689_at
Expression level (log2)
Age
Smoker
non−smoker

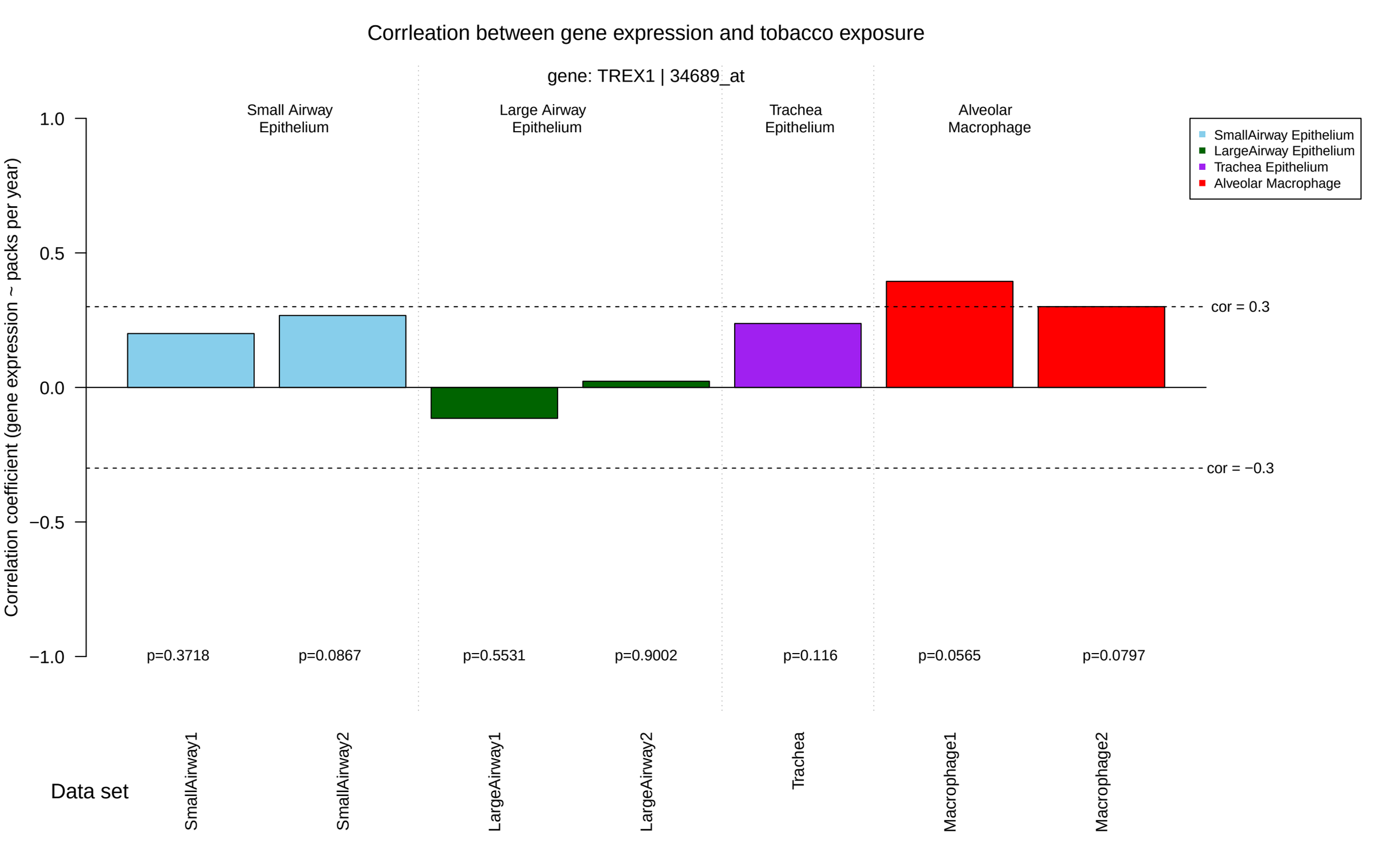

set: LargeAirway1 (p=0.9685)
TREX1 | 34689_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0165)
TREX1 | 34689_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0013)
TREX1 | 34689_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0538)
TREX1 | 34689_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.013)
TREX1 | 34689_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TREX1 | 34689_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3718
p=0.0867
p=0.5531
p=0.9002
p=0.116
p=0.0565
p=0.0797
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2